BRAF fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes BRAF gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 15 fusion events. 15/15 fusions with known frame status (100.0%) were in-frame. 9/15 fusions (60.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 100.0% in-frame and 60.0% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 15 analyzed fusion events, identifying 8 distinct fusion partner genes. The most frequent partner was SND1, observed in 5 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 9/15 (60.0%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.100899.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 8/10 (80.0%) of in-frame fusions disrupted the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.120879.

Junction concentration

62% of 8 fusions with this gene as the 3' partner join at exon 9, against 3% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; label-permutation p=0.346653). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -77 aa from the nearest configured domain boundary at 457 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=8) and "not_retained" (n=2). For Frame_status, retained: 8/8 (100.0%, 95% CI 67.6%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: BRAF: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (4 row(s) in genomic_bin_recurrence, 2 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: No jointly assessed samples for BRAF point_mutation (V600E); skipped.

Window detection

Window detection produced results for this run (8 row(s) in window_scan, 2 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SND1	5	0.3333333333333333	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22841697043154457	1
AP3B1	2	0.13333333333333333	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15782873513742693	2
FAM114A2	2	0.13333333333333333	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15782873513742693	3
MACF1	2	0.13333333333333333	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15782873513742693	4
BCL2L11	1	0.06666666666666667	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.13429932337272107	5
MKRN1	1	0.06666666666666667	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.13429932337272107	6
SUGCT	1	0.06666666666666667	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.13429932337272107	7
ZC3HAV1	1	0.06666666666666667	0.0996112703696676	0.09176487071628685	None	0.5089863313113864	confidence_certainty, domain_disruption, domain_retention, recurrence	0.13429932337272107	8

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
380	0	8	1	1	0.0	0.19999999999999998	0.6989700043360189
381	5	3	1	1	1.6666666666666667	1.0	-0.0
393	7	1	1	1	7.0	0.37777777777777777	0.4227635923970698

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
438	7	1	2	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

8	0
2	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
THPA	[8, 0], [2, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	10	2	0	8	0.0
cysteine_rich	Cysteine-rich domain	10	2	0	8	0.0

domain_retention tables

frame_domain_contingency_table

8	0
2	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
THPA	[8, 0], [2, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	10	8	0	2	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	8	10	0.8

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
AP3B1	2	2	1	1
BCL2L11	1	1	1	1
FAM114A2	2	2	1	1
MACF1	2	2	1	1
MKRN1	1	1	1	1
SND1	5	5	4	4
SUGCT	1	1	1	1
ZC3HAV1	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	140787000	140788000	5	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11
7	140783000	140784000	3	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4
7	140781000	140782000	1	EVT-TCGA-KS-A4ID-01-15
7	140794000	140795000	1	EVT-TCGA-KS-A41I-01-7

exact_position_recurrence

chromosome	position	n_events	event_ids
7	140787584	5	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11
7	140783157	2	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
381	5	1	0	False
393	2	1	0	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	9	5	8	0.625	0.034798006836058146	5.582346943857476
three_prime	10	2	8	0.25	0.023303339785034797	1.858601824306382
five_prime	8	1	2	0.5	0.035431165836181694	1.1573473071582614
five_prime	10	1	2	0.5	0.007536136391714368	1.8234607965836256
three_prime	11	1	8	0.125	0.007438331342914797	1.2367196985163837

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
181	381	200	6	4	5	3	1.6666666666666667	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11, ... (6 total)
281	381	100	6	4	5	3	1.6666666666666667	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11, ... (6 total)
331	381	50	6	4	5	3	1.6666666666666667	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11, ... (6 total)
356	381	25	6	4	5	3	1.6666666666666667	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
389	439	50	4	6	3	5	0.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-KS-A4ID-01-15
393	443	50	4	6	3	5	0.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-KS-A4ID-01-15
393	493	100	4	6	3	5	0.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-KS-A4ID-01-15
393	593	200	4	6	3	5	0.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-KS-A4ID-01-15

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
356	381	25	6	4	5	3	1.6666666666666667	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-FY-A40N-01-14, EVT-TCGA-J8-A3O1-01-11, ... (6 total)
389	439	50	4	6	3	5	0.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-KS-A4ID-01-15

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3T0-01-1	TCGA-EL-A3T0-01	TCGA-EL-A3T0	Thyroid Cancer	THPA	AP3B1-BRAF	AP3B1	in-frame	three_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | AP3B1-BRAF Fusion	in-frame
EVT-TCGA-ET-A2MX-01-2	TCGA-ET-A2MX-01	TCGA-ET-A2MX	Thyroid Cancer	THPA	BCL2L11-BRAF	BCL2L11	in-frame	three_prime	140783157	10	393	False	393	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | BCL2L11-BRAF Fusion	in-frame
EVT-TCGA-EL-A3T0-01-3	TCGA-EL-A3T0-01	TCGA-EL-A3T0	Thyroid Cancer	THPA	BRAF-AP3B1	AP3B1	in-frame	five_prime	140794308	8	380	False	380	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | BRAF-AP3B1 Fusion	in-frame
EVT-TCGA-ET-A3BN-01-4	TCGA-ET-A3BN-01	TCGA-ET-A3BN	Thyroid Cancer	THPA	BRAF-FAM114A2	FAM114A2	in-frame	five_prime	140783021	10	438	False	438	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | BRAF-FAM114A2 Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-5	TCGA-DE-A0Y2-01	TCGA-DE-A0Y2	Thyroid Cancer	THPA	BRAF-MACF1	MACF1	in-frame	five_prime	140794308	8	380	False	380	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | BRAF-MACF1 Fusion	in-frame

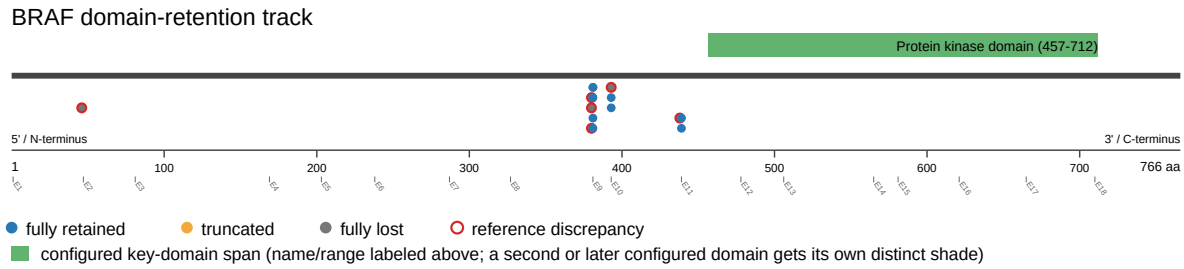
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3ZK-01-6	TCGA-EL-A3ZK-01	TCGA-EL-A3ZK	Thyroid Cancer	THPA	BRAF-SND1	SND1	in-frame	five_prime	140787548	9	393	False	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False														RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | BRAF-SND1 Fusion	in-frame
EVT-TCGA-KS-A41I-01-7	TCGA-KS-A41I-01	TCGA-KS-A41I	Thyroid Cancer	THPA	BRAF-SND1	SND1	in-frame	five_prime	140794308	8	380	False	380	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | BRAF-SND1 Fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-FY-A40N-01-8	TCGA-FY-A40N-01	TCGA-FY-A40N	Thyroid Cancer	THPA	BRAF-SUGCT	SUGCT	in-frame	five_prime	140924566	1	46	False	46	True	exact match	lost	0.0	False													Protein kinase domain; RAS-binding domain; Cysteine-rich domain			NA | BRAF-SUGCT Fusion	in-frame
EVT-TCGA-ET-A3BN-01-9	TCGA-ET-A3BN-01	TCGA-ET-A3BN	Thyroid Cancer	THPA	FAM114A2-BRAF	FAM114A2	in-frame	three_prime	140781693	11	439	False	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | FAM114A2-BRAF Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-10	TCGA-DE-A0Y2-01	TCGA-DE-A0Y2	Thyroid Cancer	THPA	MACF1-BRAF	MACF1	in-frame	three_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | MACF1-BRAF Fusion	in-frame
EVT-TCGA-J8-A3O1-01-11	TCGA-J8-A3O1-01	TCGA-J8-A3O1	Thyroid Cancer	THPA	MKRN1-BRAF	MKRN1	in-frame	three_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | MKRN1-BRAF Fusion	in-frame
EVT-TCGA-DJ-A3UZ-01-12	TCGA-DJ-A3UZ-01	TCGA-DJ-A3UZ	Thyroid Cancer	THPA	SND1-BRAF	SND1	in-frame	three_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | SND1-BRAF Fusion	in-frame

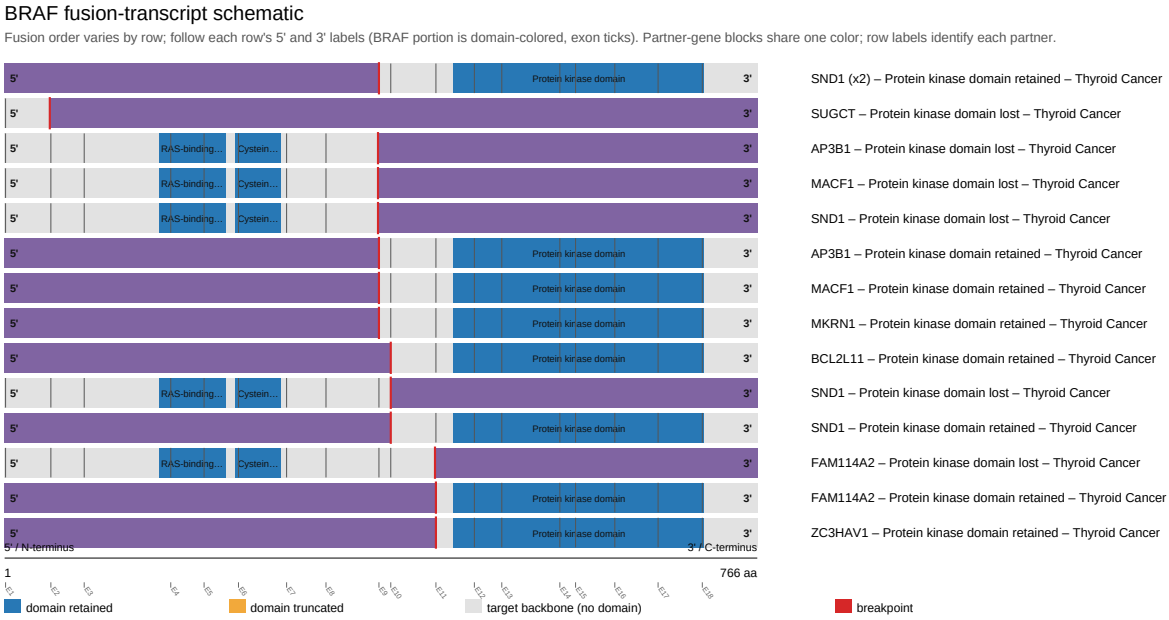
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crossscheck_protein_position	fusion_annotation_crossscheck_agrees	fusion_annotation_crossscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3ZK-01-13	TCGA-EL-A3ZK-01	TCGA-EL-A3ZK	Thyroid Cancer	THPA	SND1-BRAF	SND1	in-frame	three_prime	140783157	10	393	False	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | SND1-BRAF Fusion	in-frame
EVT-TCGA-FY-A40N-01-14	TCGA-FY-A40N-01	TCGA-FY-A40N	Thyroid Cancer	THPA	SND1-BRAF	SND1	in-frame	three_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | SND1-BRAF Fusion	in-frame
EVT-TCGA-KS-A4ID-01-15	TCGA-KS-A4ID-01	TCGA-KS-A4ID	Thyroid Cancer	THPA	ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140781693	11	439	False	439	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | ZC3HAV1-BRAF Fusion	in-frame

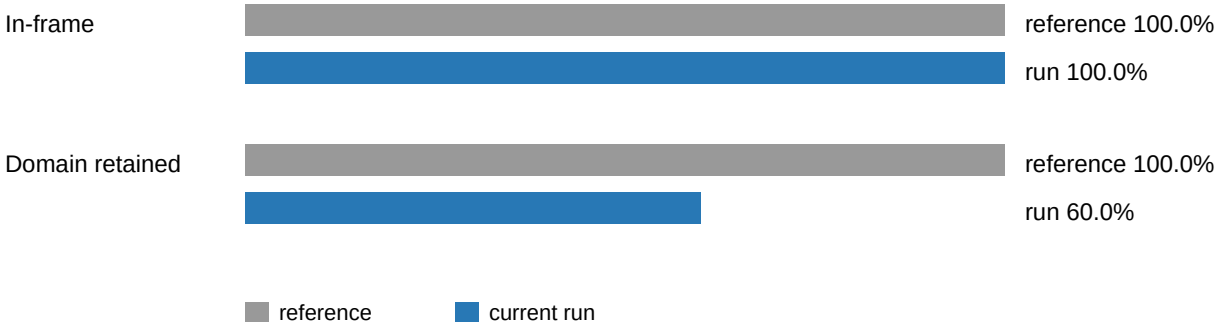
Figures

domain retention outliers



fusion schematic



reference comparison

Reference vs thca_tcga_pan_can_atlas_2018